

How much story can you get from one eye?

Ageing and other cool stuff from shark eyes

Old Sharks



The eyes
have it

NPRB 2301: Ageing of Pacific Sleeper Shark



Cindy Tribuzio, AFSC ABL

Charlotte Springer, UAF/AKBioMap

Beth Matta, AFSC Age and Growth

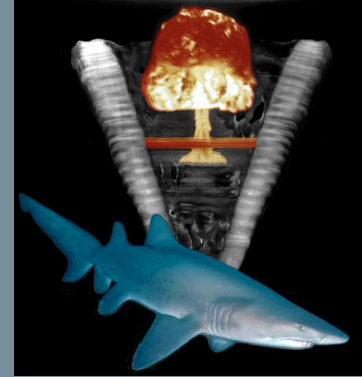
Darcie Neff, AKBioMap

Taylor Broek, Stanford

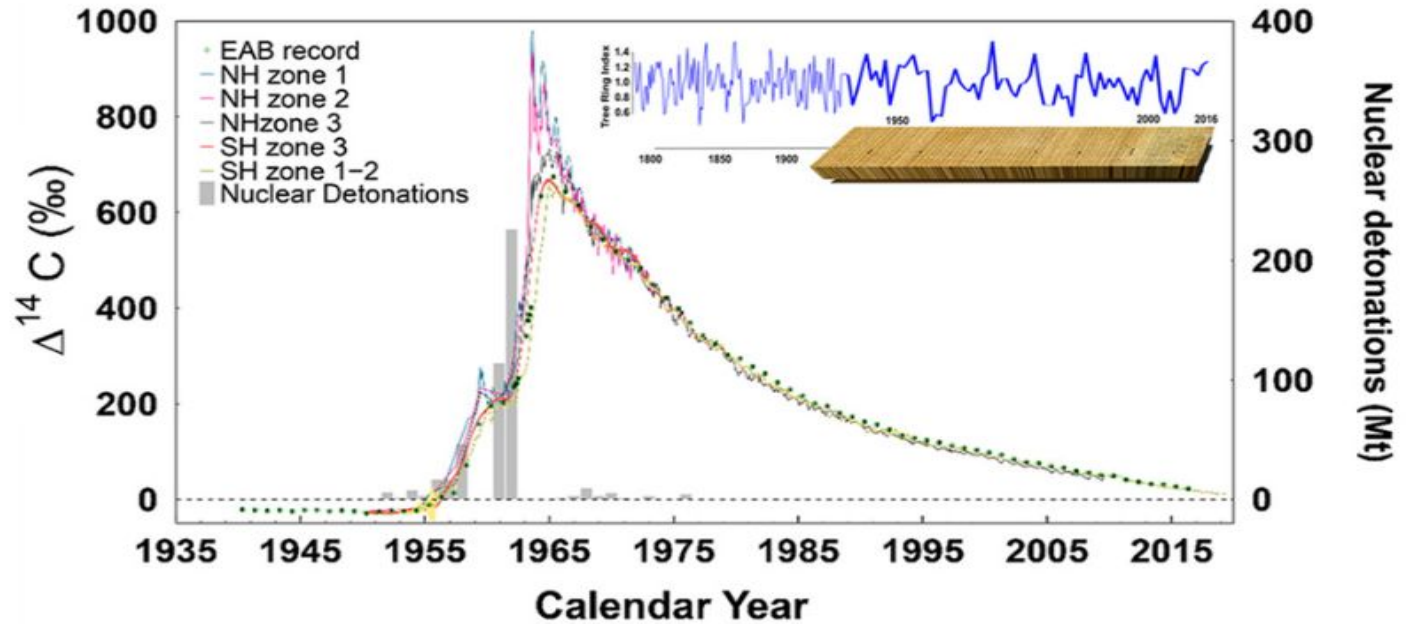
Allen Andrews, Age and Longevity Research

Bruce Buchholz, Lawrence Livermore National
Laboratories

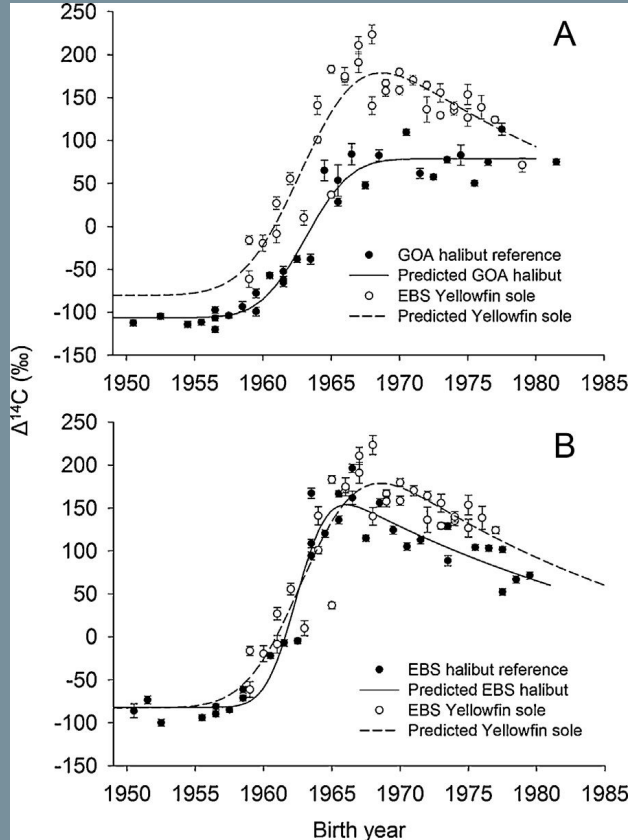
Dan Stewart, American River College



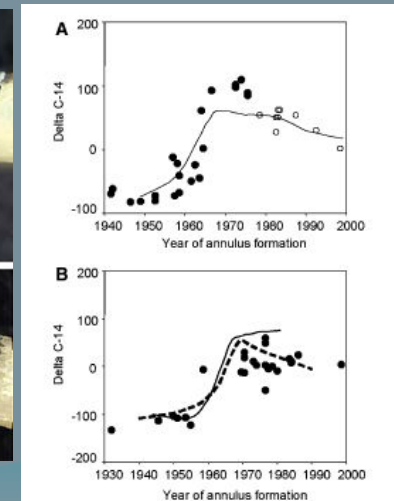
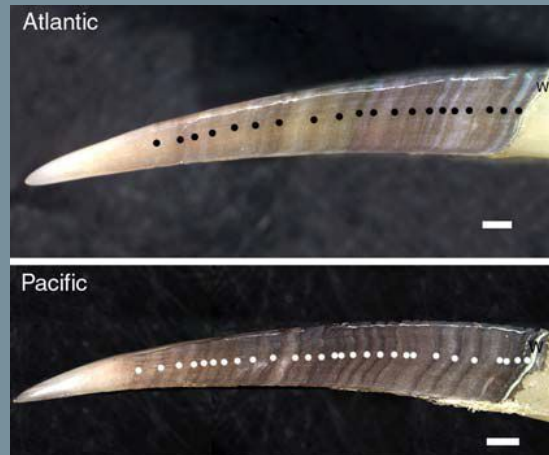
Bomb Radiocarbon Dating



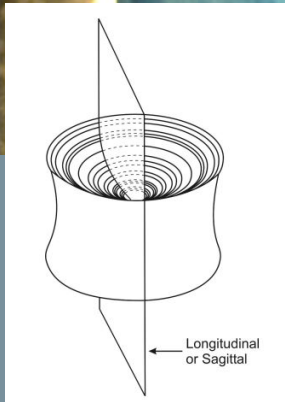
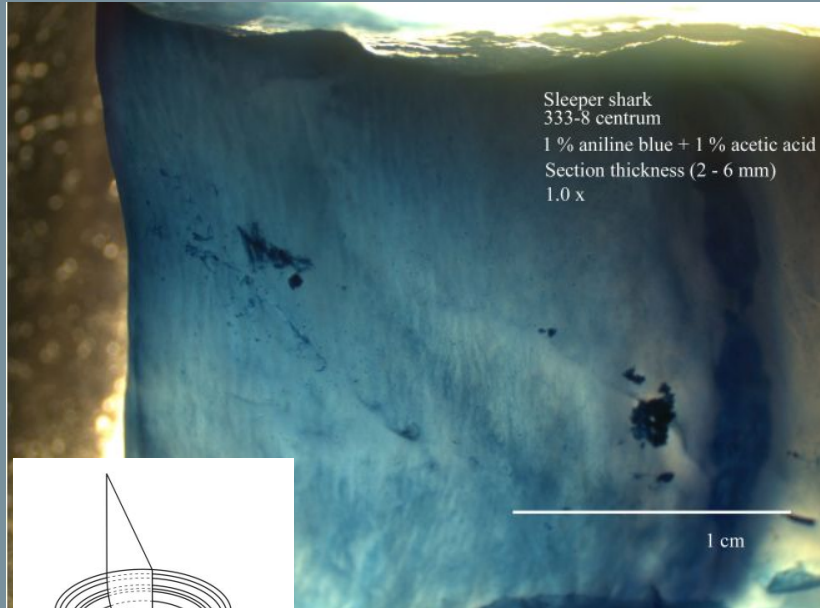
Bomb Radiocarbon Dating



- Often used in teleost and elasmobranchs to validate ageing methods
- Applied to species which HAVE ageing methods



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Bomb Radiocarbon Dating



- Often used in teleost and elasmobranchs to validate ageing methods
- Applied to species which HAVE ageing methods
- Attempts at ageing Somniosids (aka sleeper sharks) with traditional methods have all failed
- Using bomb radiocarbon methods to establish longevity

Bomb Radiocarbon

Are pet clinical trials ready
for prime time? p. 638

The sense (and antisense)
of ALS therapy pp. 647 & 708

Keeping fam
their homes

Science

\$15
12 AUGUST
sciencemag.org

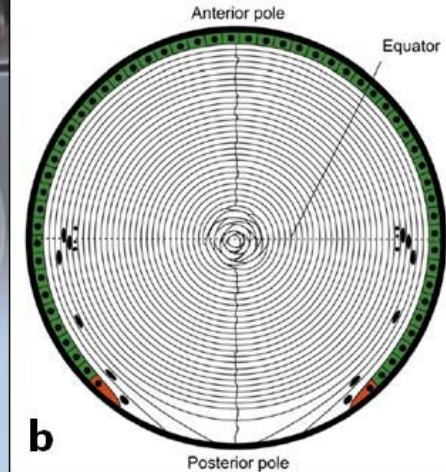


LONG IN THE TOOTH

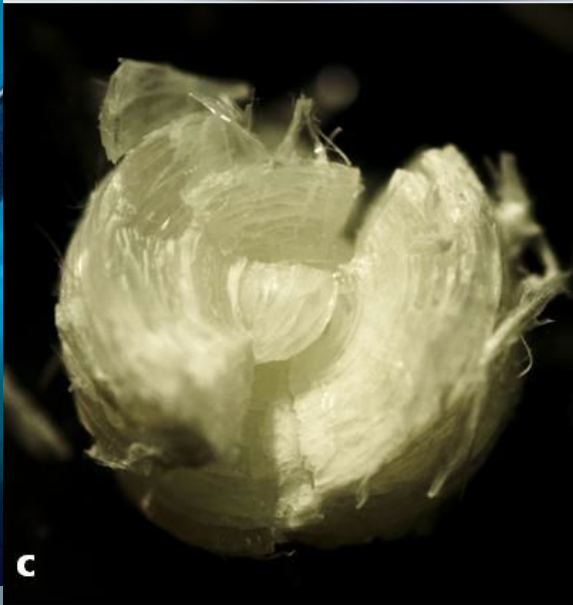
400-year-old sharks in
Arctic waters p. 702



a



b



c



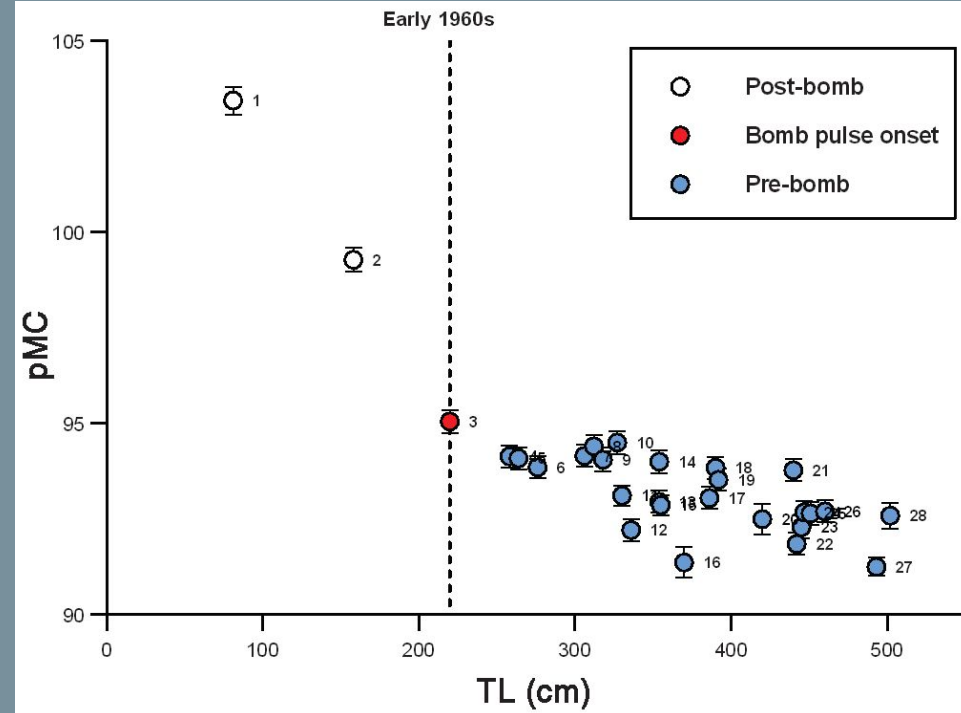
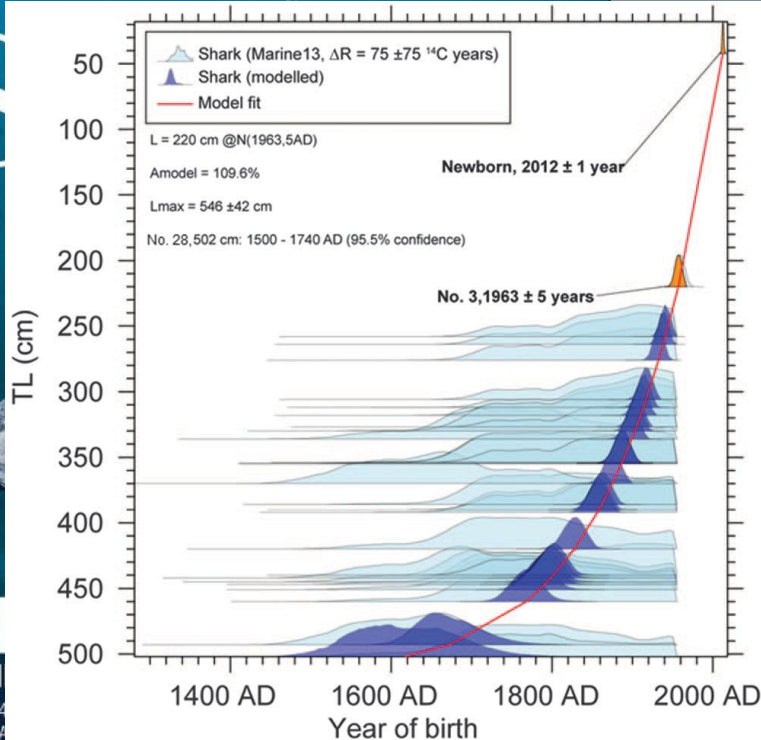
d

Bomb Radiocarbon Dating

Are pet clinical trials ready
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The sense (and antisense)
of ALS therapy pp. 647 & 708

Keeping families in
their homes p. 694



NPRB 2301: Ageing of Pacific Sleeper Shark

157 Pacific sleeper sharks
sampled
40 - 465 cm TL
Each eye layer retained for
chemical analysis
Spiny dogfish with pups for
comparison

Differences:

Nielsen et al. 2016

- Dissected lens at sea
- Dried whole
- Only core analyzed for delta C14
- Small sample size

NPRB 2301

- Frozen at sea
- Delaminated wet
- All layers analyzed for C14, stable isotopes (C13 and N15), CSIAA
- Large sample size with many samples per animal

For a great overview of the species:

[Matta et al. 2023 Western Groundfish Conference.](#)

[Matta et al. 2024. Polar Biology article](#)



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Peeling eye layers



465 cm TL Female



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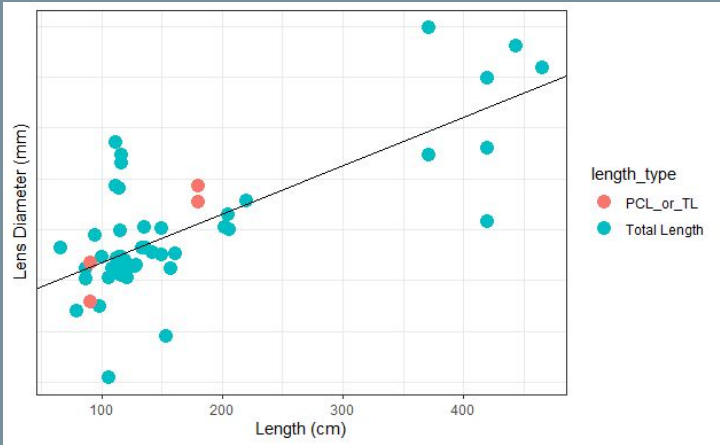
Peeling eye layers



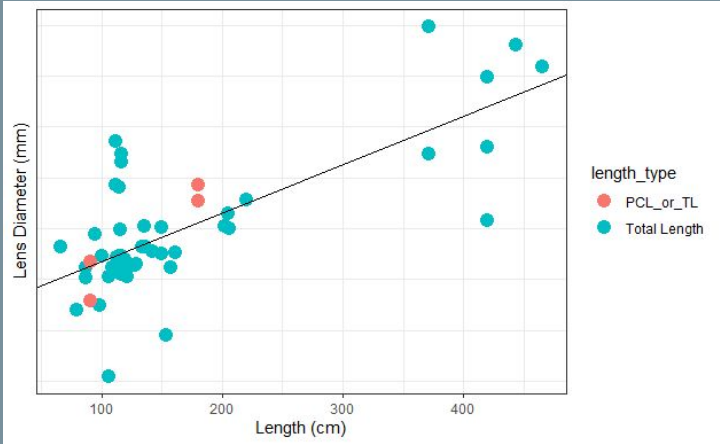
465cm TL Female
Pending results



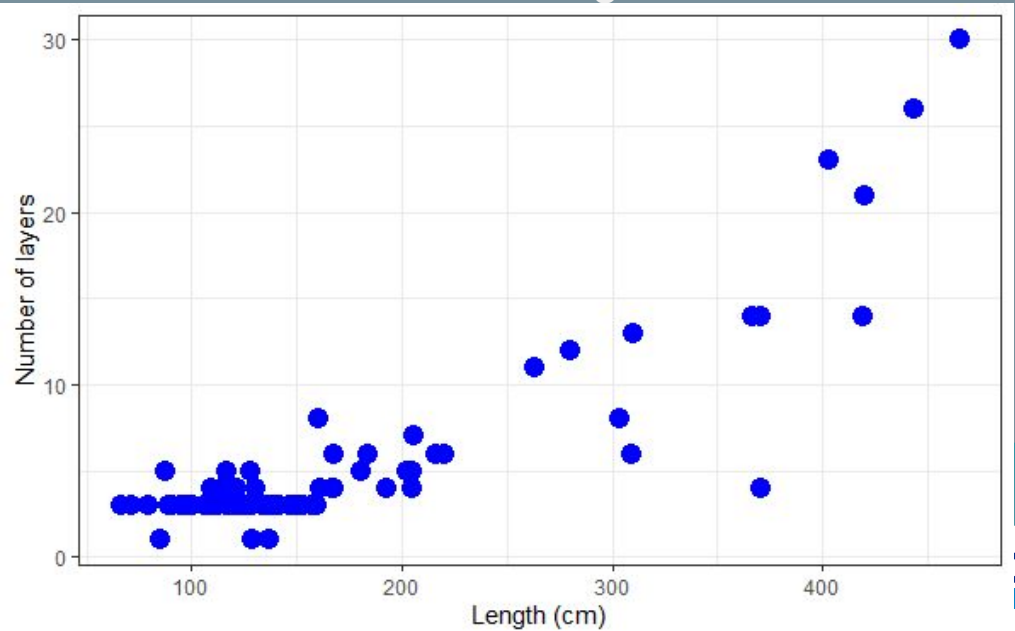
NPRB 2301: VERY preliminary results



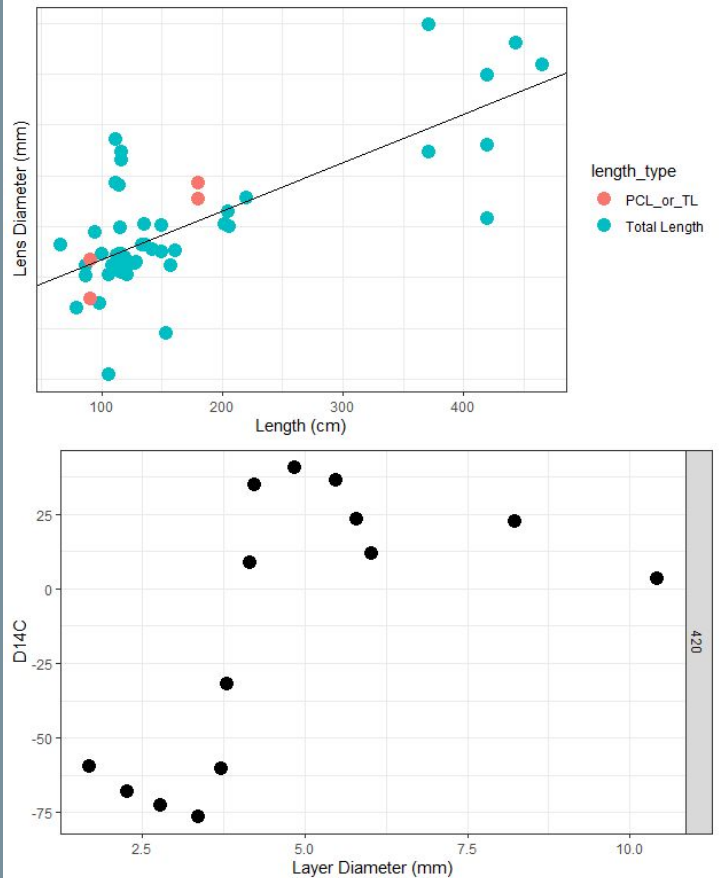
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Assume linear relationship between eye lens diameter and total length



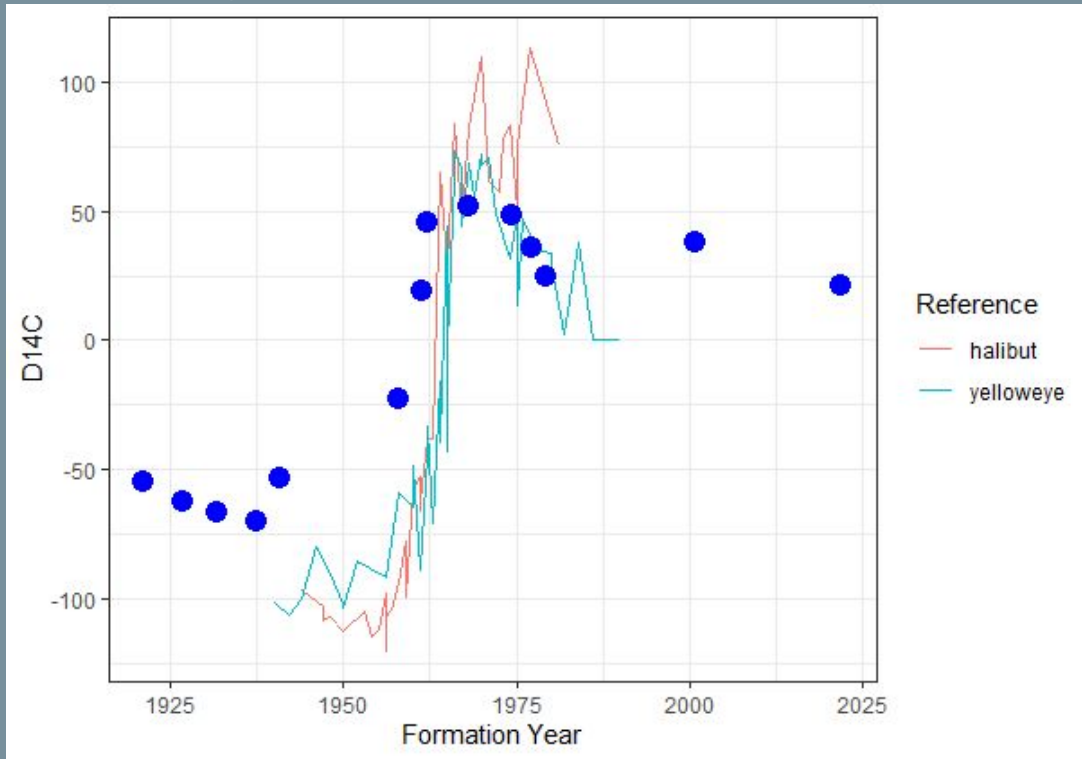
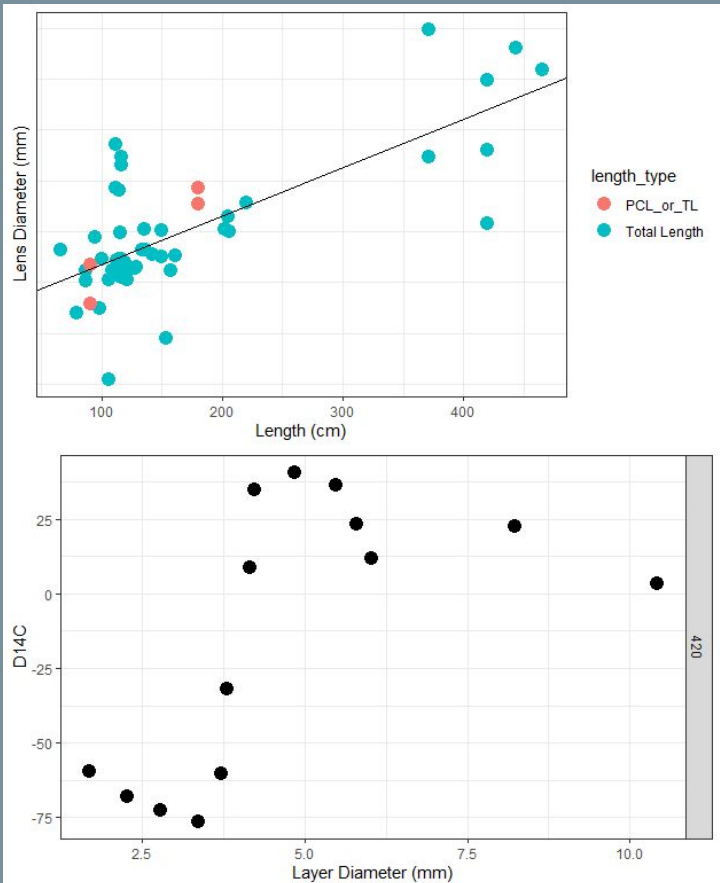
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420 cm TL, GOA, 2022

NPRB 2301: VERY preliminary results



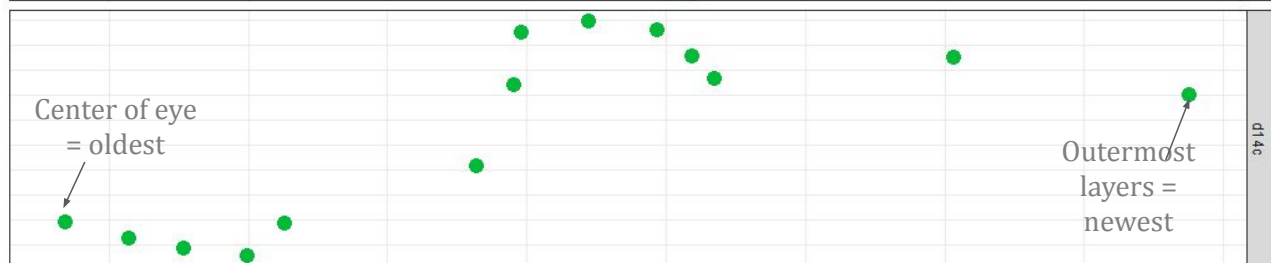
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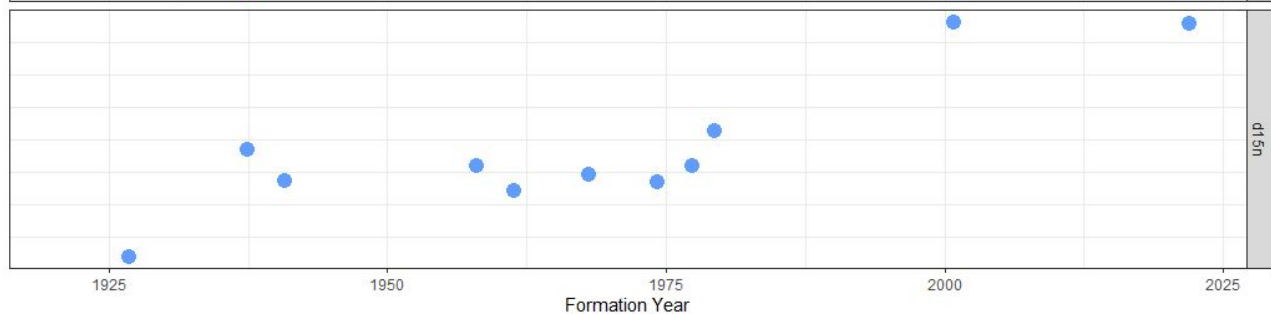
13C indicator
habitat or region



14C indicator
of age



15N indicator
trophic position



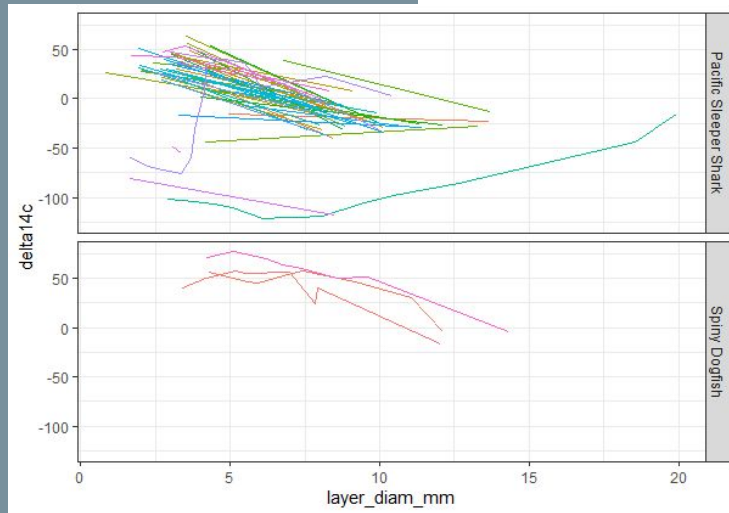
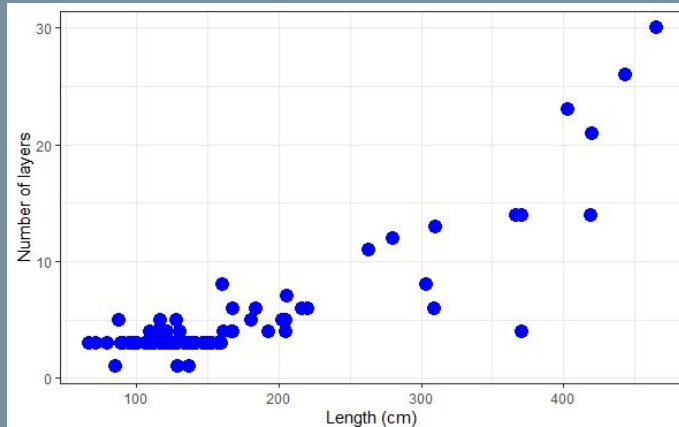
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NPRB 2301: VERY preliminary results

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- Some sample sleuthing needed
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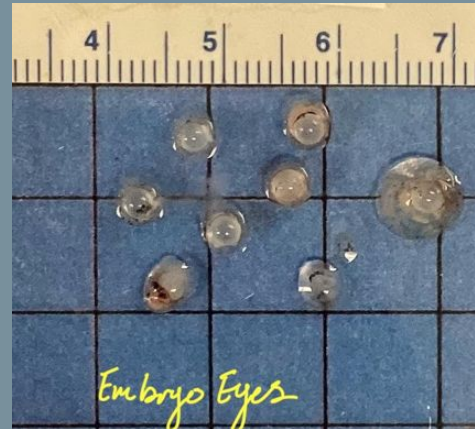
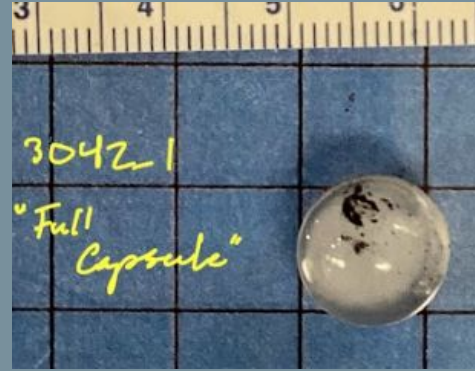


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NPRB2301: other analytics pending results

Spiny dogfish

- Maternal/offspring fractionation
- Bonus 14C/SIA of high resolution dogfish layers



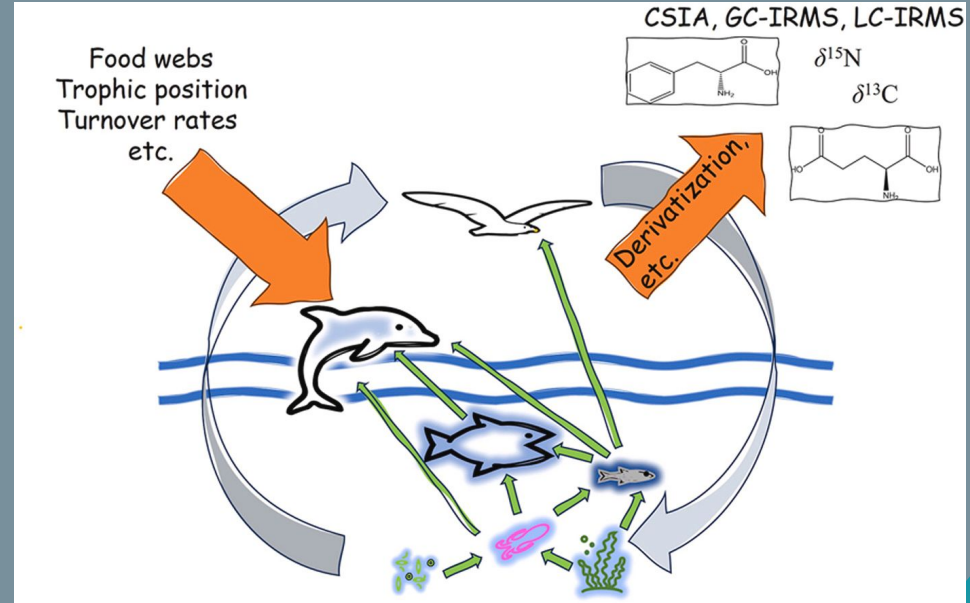
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CSIAA-A

- Both species



Khaliq et al. 2024

NPRB2301: other analytics pending results

Spiny dogfish

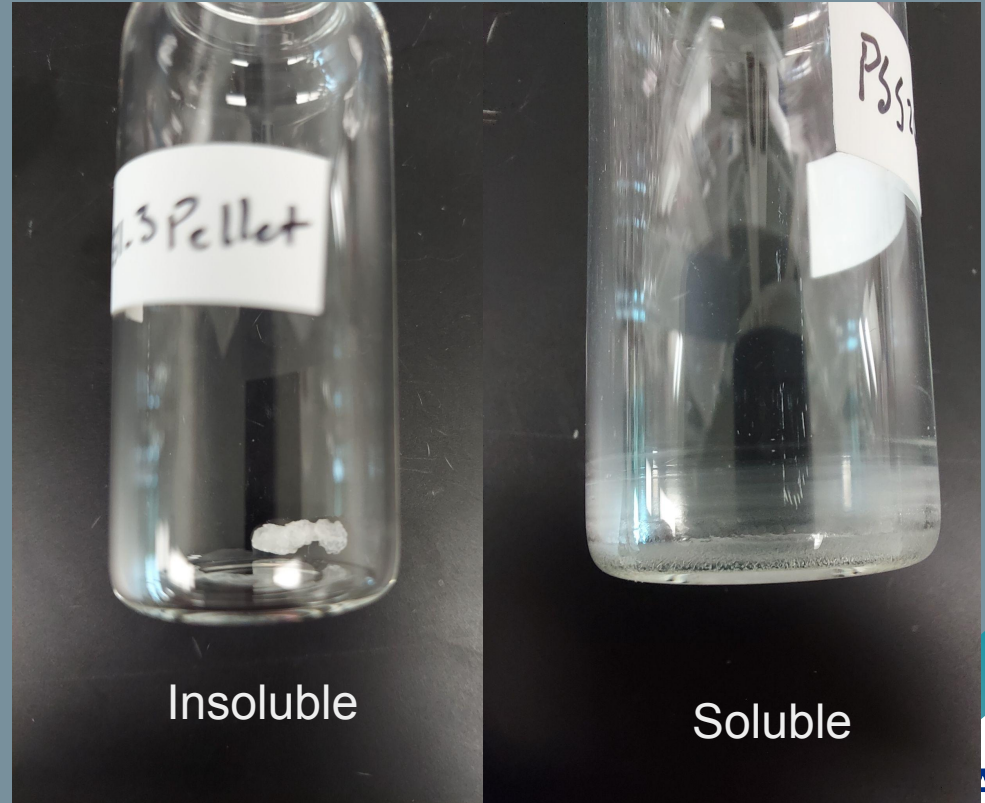
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Carbon turnover

- ^{14}C in soluble vs insoluble proteins



NPRB2301: end goals

December 2022

BSAI Sharks

19. Assessment of the Shark Stock Complex in the Bering Sea/Aleutian Islands and Gulf of Alaska

Cindy A. Tribuzio, Mary Elizabeth Matta, Katy B. Echave, Cara Rodgveller,
Garrett Dunne and Keith Fuller

November 2022

EXECUTIVE SUMMARY

This document presents the assessment for the shark stock complex (Pacific spiny dogfish, Pacific sleeper shark, salmon shark and other/unidentified sharks) in both the Gulf of Alaska (GOA) and Bering Sea/Aleutian Islands (BSAI) Fishery Management Plan (FMP) areas. While advice remains separate by FMP, recent tagging and genetic studies suggest that the stocks are shared between these areas. We combined the assessments here to streamline the presentation of data that are in common (e.g., life history, data summaries, etc.) and to harmonize advice and management recommendations between regions.

For the BSAI, the SSC has placed this complex within Tier 6 of the FMP. This means the OFL is based on the maximum historical catch between the years 2003-2015, and the ABC is 75% of OFL. For the GOA, the complex is managed as a combination of Tier 5 (for spiny dogfish) and Tier 6 species (for all other sharks). The OFL and ABC for the GOA complex sums over these tiers. The GOA spiny dogfish assessment uses the random effects while the remaining components (Tier 6) are based on species-specific average catches from 1997-2007.

BSAI - Summary of Changes in Assessment Inputs

Changes to the input data

1. Total catch for BSAI sharks is updated for 2003-2022 (as of Oct 8, 2022)

Changes in assessment methodology

We proposed changes for how 3 of the 4 BSAI shark species categories are treated. For Pacific sleeper sharks, we evaluated stock status based on a catch-only model; for the other two, we illustrate results from a modified Tier 6 approach where the 90th percentile of the catch series is used instead of the maximum. The ABC/OFL recommendation from the salmon shark component of the complex would remain the same.

BSAI - Summary of Results

There is no evidence to suggest that overfishing is occurring for any shark species in the BSAI because the OFL has not been exceeded. Total shark catch in 2021 was 221 t, and catch in 2022 was 123 t, as of October 8, 2022. On average, 12% of the total annual catch occurs after October 1st each year.

For 2023-2024 we **recommend the maximum allowable ABC of 293 t and an OFL of 391 t for the shark stock complex in the BSAI**. The recommended ABC/OFL are a 43% decrease from the previous assessment and are due to the recommended model changes. Current catches are below the recommended ABC and have been below the recommended ABC over the last 15 years. It is unlikely that shark catches would increase such that the ABC would be reached. There are currently no directed commercial fisheries for shark species in the BSAI, and most incidental catch is discarded.

Advance the stock assessment by informing:

- Estimates of longevity
- Estimates of growth rate
- Natural mortality estimation
- Exploration of expanded data-limited assessment methods
- Demographic simulations



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Thanks!!!

North Pacific Research Board for funding and next level support to solve logistical problems

Emily F, Cody P, Matt R and Bryan C for all the help and patience

MESA staff for listening to my ramblings and complaints

Picture: Clark et al. 1990
Sugura Bay, Japan
Est size: 660-720 cm TL
Matta et al. 2024



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